SerialPortManager

SerialPortManager Quick Guide

Version 0.4.0 - Edition 2026-10-14

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Version 0.3.0 - SerialPortManager (SerialPM) for Windows.

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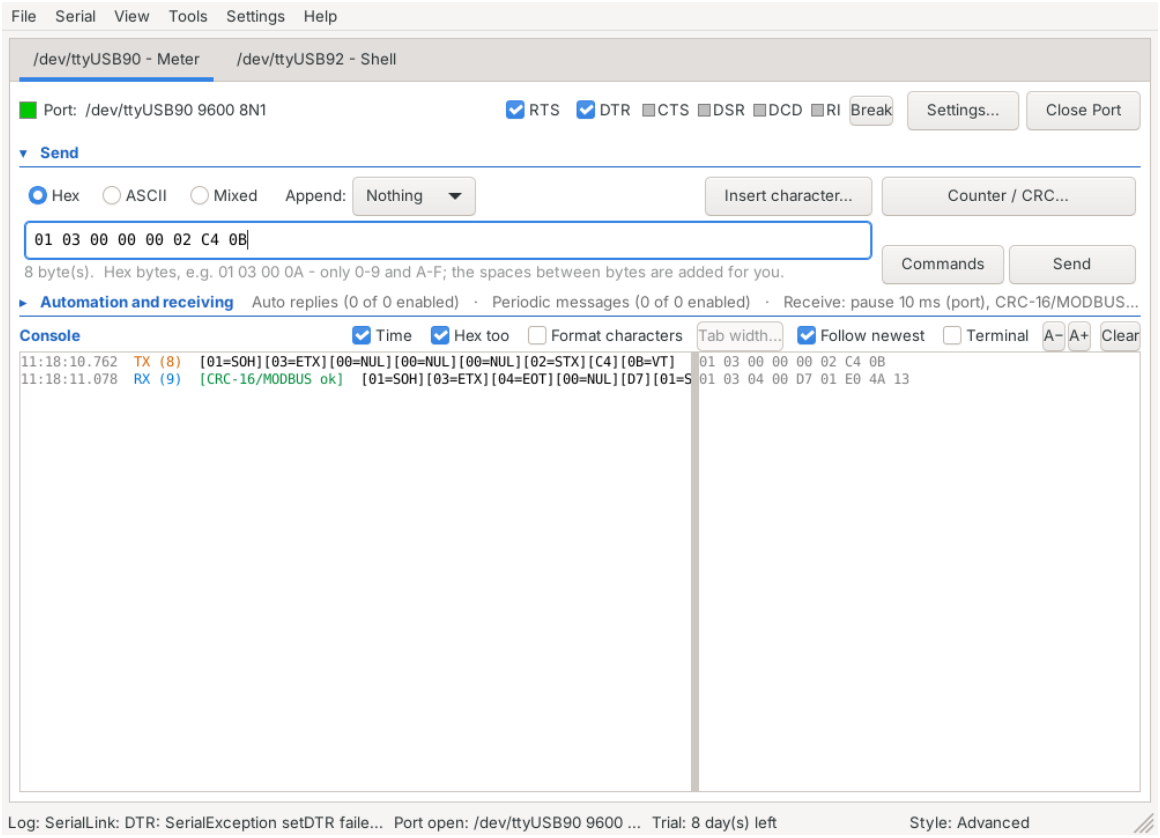
1. What it is

SerialPortManager (SerialPM) is a Windows tool for serial ports: COM ports, USB-serial adapters, RS-232 / RS-422 / RS-485 converters. It shows everything received and sent, sends data typed in hexadecimal or text, replies automatically, sends periodic messages, forwards data between ports, transfers files, and analyses bytes: checksums and CRCs, numbers in every byte order, Modbus and NMEA frames, and the waveform on the wires.

Several ports work at the same time: each one has its own tab with its own port, console and automation. This guide covers the essentials; the **User Manual** (**Help > User Manual**, F1) describes every feature.

2. Install and first start

1. Copy the program folder to a place where you can write (not *Program Files*) and start **SerialPortManager.exe**. No installer is needed.
2. At the first start the program creates the folder **SerialPortManager data** next to the exe. It holds the settings (AppSettings.txt), the event database (events.db), the log (log.txt), the language files and the license. **Settings > Open Data Folder** opens it.
3. The 14-day trial starts: every feature is available. The status bar shows **Trial: N day(s) left**.
4. Optional: **Settings > Language...** (English / Italiano) and **Settings > Theme...** (Light / Dark). The program asks **Restart now**, **Use now**, **Save for next start** or **Cancel**.
5. Optional: **Settings > User Preferences...** - how much the window shows: **Simple** (green: port, Send, console), **Advanced** (blue, the default), **Pro** (purple) or **Full** (orange), or feature by feature; **Settings > Text Size...** - the size of the text of the windows, and apart of the console and the terminal.



The main window

The main window has the menus **File**, **Serial**, **View**, **Tools**, **Settings**, **Help**; one tab per port, with the sections **Send**, **Automation and receiving** and **Files, capture and replay** that open and close with a click on their title (closed, they show a summary), and the console; the application log under the tabs with **View > Application Log**; a status bar with messages, the port state of the selected tab, the license state and the style.

3. Open a port

1. Choose **Serial > Serial Ports...** to set **How many serial ports to control:** (one tab each, at most the number of ports present) and a **Name:** for each; **Configure...** sets the port of each slot. For one tab only, click **Settings...** in the tab.
2. In the **Serial Port** dialog choose **Port** (**Refresh** reads the list again), **Baud rate**, **Data bits**, **Parity**, **Stop bits**, **Flow control** and the timeouts. The default is 9600 8N1, no flow control.
3. Click **Open Port**. The LED turns green and the button becomes **Close Port**.

A port belongs to one tab only: a port used elsewhere is listed as **(already used by Slot N)**; choosing it asks **Detach and use here** or **Abort**.

Inter-byte timeout (ms) decides when a received message ends: bytes are grouped into one console line until the line is quiet for that time. Increase it if messages appear split, decrease it if several messages appear merged.

Receive: Edit... in the tab cuts messages differently - by a delimiter (CR LF), a fixed length, a length field or Modbus RTU timing - and can check the checksum / CRC of every received message (**[CRC-16/MODBUS ok]** green, **WRONG** red).

The port row also has the control lines: **RTS** and **DTR** (outputs, remembered), the LEDs **CTS**, **DSR**, **DCD**, **RI** (inputs) and **Break** (250 ms). If a USB adapter is unplugged, the port is closed and opened again when it comes back (orange LED, **Stop Waiting**). The port list shows what each device is, for example *COM3*

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(USB-SERIAL CH340).

Network... in the port settings chooses a port over the network: `tcp://host:port` (raw), `tcpserver://:port` (waits for a client) or `rfc2217://host:port` (the speed and the lines set remotely). Framing, parity and overrun errors and breaks received are noted in the console (**Lines**).

4. Send

The **Send** box has three notations:

Notation	Example
Hex	`01 03 00 0A` (spaces are added for you)
ASCII	`HELLO` (printable characters only)
Mixed	`OK//0x0D//0x0A` (text, and `//0x` + two hex digits for any byte)

Type the data and press **Enter** or click **Send**. Clicking another notation converts the text; the bytes stay the same. **Insert character...** opens a table of all 256 byte values: double-click adds a character and closes the table, right-click adds it and keeps the table open. A mistake is shown in red under the text, for example **Character 7: only hex digits (0-9, A-F) are allowed.**

Append: adds **CR**, **LF** or **CR and LF** to every message sent, without showing it in the text. **Counter / CRC...** inserts an incrementing counter (1-8 bytes, step per send, little- or big-endian) and a checksum / CRC (any algorithm of the Byte Analyzer), each at the place you choose: last, before the last byte, first... - the dialog previews the next message, for example `01 03 00 00 00 01 [07] {4B A1} 0D 0A`.

Up / **Down** in the text bring back the messages sent. **Commands** next to **Send**: saved commands sent with one click, **Save the text as a command...**, and the history.

Commands > Show the command buttons (Shift+F1...F12): a button per saved command under the send box; Shift+F1 ... Shift+F12 send the first twelve.

5. Read the console

Every message is one line of the console:

```
14:03:21.512  RX (5)  HELLO        48 45 4C 4C 4F
```

Time, label and byte count and the bytes as text (control bytes as `[0D=CR]`, bytes above 7E as `[HH]`) on the left; the same bytes in hex in the right pane, line for line - the two panes scroll together. Labels and colours: **RX** blue (received), **TX** orange (sent by you), **TX auto "name"** purple (auto reply), **TX periodic "name"** green, **TX fwd "tab"** teal (forwarded), file transfer lines amber, errors red.

Options above the console: **Time**, **Hex too** (the hex pane), **Format characters** (CR, LF and CR LF start a new line, tabs become spaces - **Tab width...**), **Follow newest** (both panes; turn it off to read older lines while data arrives) and **Clear**.

Terminal shows a VT100 / ANSI terminal instead of the console: keys go to the device as you type them, with the device's colours and cursor (shells, boot loaders, menus); right-click for copy, paste, local echo and what Enter and Backspace send. **A−** / **A+** (or Ctrl + mouse wheel) make the text of the console and the terminal smaller or larger. **Decode Modbus** (right-click) shows the meaning of Modbus frames next to every message.

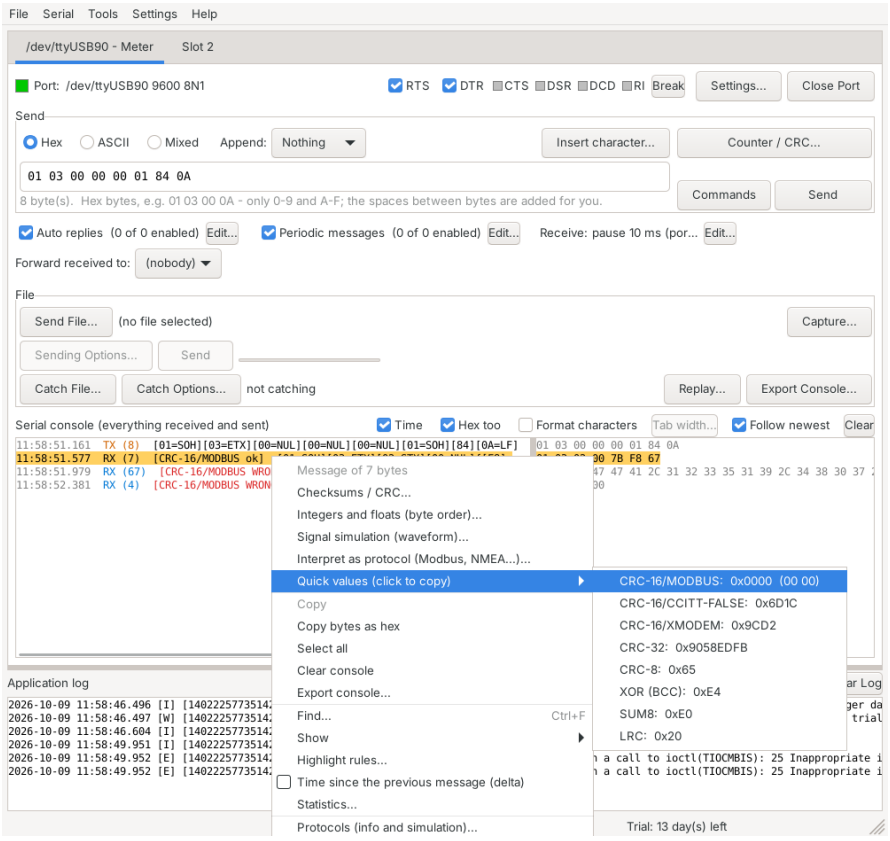
Right-click menu of the console, besides the analysis: **Export console...** (text or CSV), **Find...** (Ctrl-F: find bar, **Only matching lines**), **Show** (received / sent / notes), **Highlight rules...** (a text or bytes on a colour), **Time since the previous message (delta)**, **Statistics...**. In **Files, capture and replay**: **Capture...** the traffic to a file (`*.spmcap` with times, or raw `*.bin`), **Replay...** a capture, **Export Console...**.

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6. Auto replies, periodic messages, forwarding, files

- **Auto replies**: **Edit...** next to the switch (or **Serial > Auto Replies...**), **Add...**: **When these bytes are received**, **Reply with**, **Delay before replying (ms):**. Tick the rule and the **Auto replies** switch.
- **Periodic messages**: **Edit...** next to the switch, **Add...**: **Send every (ms):** and the **Message**. Tick it and the **Periodic messages** switch.
- Both messages can contain **incrementing values** (counters written over the bytes at their index) and a **checksum** (SUM, XOR, LRC, CRC-8, CRC-16/MODBUS, CRC-32...) inserted where you choose. The preview shows the first messages.
- **Forward received to:** sends everything this port receives out of another tab's port.
- **File**: **Send File...**, **Sending Options...** (chunk size; pause after each chunk, or burst mode waiting for a confirmation; start prefix, end marker, recovery), **Send**. On the receiving side **Catch File...** writes the received file to the output file. **Use the recommended value** on every command makes two SerialPortManager installations understand each other.
- **XMODEM / YMODEM**: choose the protocol, **Send File...** or **Receive File...** - for boot loaders and terminal programs (`loady`, `rx`, `rb`).
- **Record into files**: everything the port receives and sends into log files - a new file at a size and every day, the oldest deleted (**Log Options...**) - for days or weeks.

7. Right-click a message: analysis



The console right-click menu

Right-click a message in the console - or select some of its bytes, in the text or in the hex part, also across lines, and right-click the selection. The menu shows **Message of N bytes** or **N selected bytes**, then:

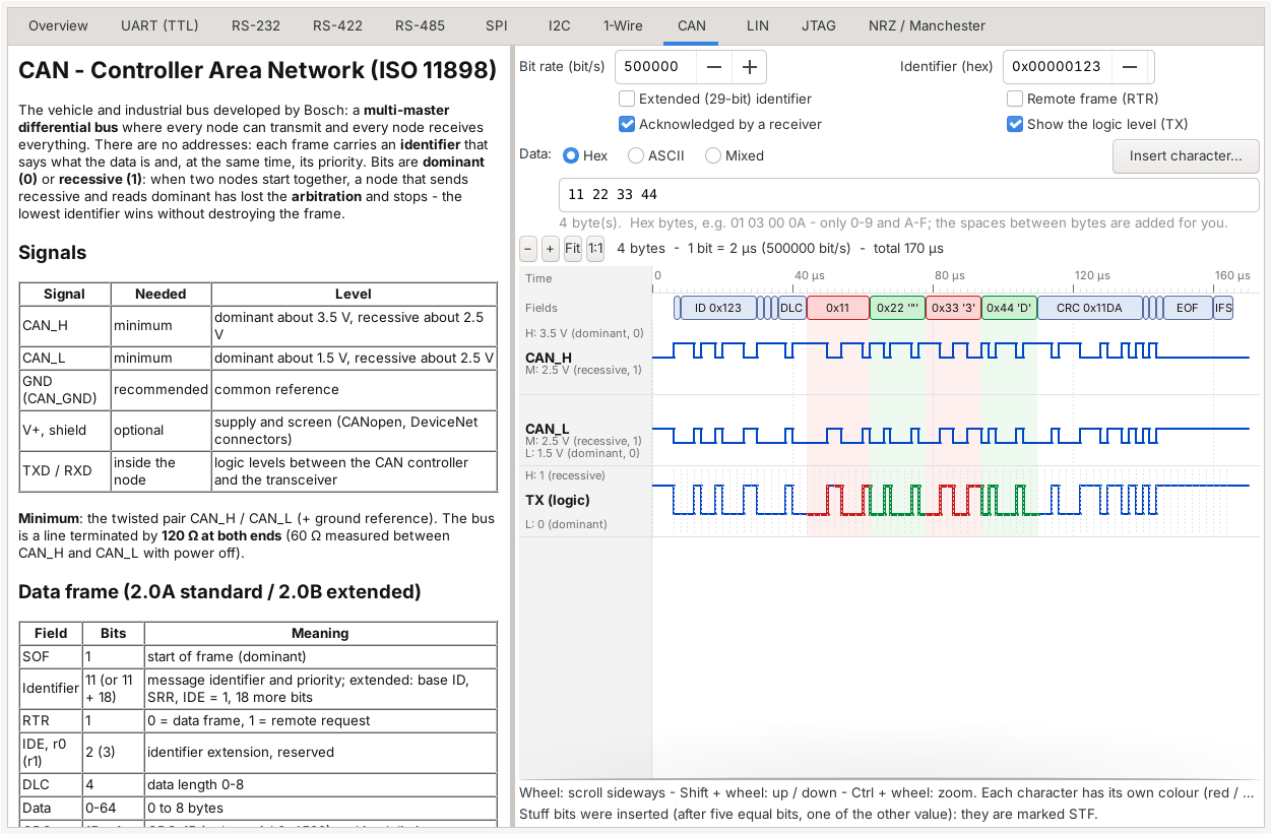
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- **Checksums / CRC...**, **Integers and floats (byte order)...**, **Signal simulation (waveform)...**, **Interpret as protocol (Modbus, NMEA...)...**: each opens a **Byte Analyzer** on that page;
- **Quick values (click to copy)**: CRC-16/MODBUS (with its two bytes, low byte first), CRC-16/CCITT-FALSE, CRC-16/XMODEM, CRC-32, CRC-8, XOR, SUM8, LRC; for 2, 4 or 8 bytes also the integer and float values in BE, LE and CDAB order. Click one to copy it;
- **Copy**, **Copy bytes as hex**, **Select all**, **Clear console**, **Protocols (info and simulation)...**.

Tools > Byte Analyzer... (Ctrl-B) opens an analyzer on the bytes of the send box. Several analyzers can be open at once; the bytes at the top can be edited. Its pages:

- **Checksums / CRC**: about 50 sums and CRCs with their published parameters, the bytes to append (big-endian and little-endian), **Message ends with it** (green when the message already ends with that checksum - this finds the checksum a device uses), **Used by**, and **Any CRC (Rocksoft parameters)**.
- **Integers / floats**: every type from Int8 to UInt64, Int24, Float16/32/64 and BCD at an **Offset**, in byte order ABCD, DCBA, BADC or CDAB, with **Scale x**; and **Value to bytes**, which finds a value in the message.
- **Signal**: the waveform of the bytes, starting from the port's baud rate, data bits, parity and stop bits.
- **Protocol**: **Automatic**, **Modbus RTU**, **Modbus ASCII**, **NMEA 0183** or **Bytes one by one**; each field with its byte range, correct checks in green, wrong ones in red.

8. The Protocols window



The Protocols window, CAN tab

Tools > Protocols (info and simulation)... (Ctrl-P) opens an independent window, **Standard communication protocols - information and simulation**. The **Overview** tab compares the protocols; one tab each explains and simulates **UART (TTL)**, **RS-232**, **RS-422**, **RS-485**, **SPI**, **I2C**, **1-Wire**, **CAN**, **LIN**, **JTAG** and **NRZ / Manchester**:

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- left: signals needed (minimum and maximum), levels, speeds, distances, frame, error checks;
- right: **Data:** and every parameter of the protocol, and the live waveform.

The waveform shows a time ruler, a **Fields** row (each character in a box, alternately red and green; protocol fields in blue), one row per wire with the meaning of its levels, 0 / 1 over every bit and the role of protocol bits under it (ST, SP, PAR, ACK, NAK, STF...). The wheel scrolls sideways, Shift + wheel up and down, Ctrl + wheel zooms at the mouse; the buttons minus, **+**, **Fit** and **1:1** set the zoom. The line under the view tells time, field, level and bit under the mouse.

9. More tools

- **Tools > Modbus Master...** (Ctrl-M): reads and writes coils and registers of a Modbus RTU / ASCII slave through a tab's port - functions 01-06, 15, 16, once or **Poll every N ms**; values as 16 / 32-bit integers or floats; a log with answers, exceptions and timeouts.
- **Tools > Modbus Slave Simulator...**: a tab's port answers as a Modbus slave from an editable table of coils and registers - to test a master without the device.
- **Tools > Line Monitor (two ports)...**: with a sniffer cable, what two devices send to each other in one list, with directions, gaps, hex, text and Modbus decoding.
- **Tools > GPS / GNSS...**: a GPS receiver on a tab's port - fix, position, speed, altitude, compass, satellites (sky plot, signal), track map with GPX / KML / CSV export, graphs; control in its own commands (MediaTek PMTK, Airoha PAIR, u-blox UBX, SiRF, SkyTraq, CASIC, Garmin): restarts, rate, baud rate, sentences, constellations.
- **Tools > Protocol Definitions...**: your device's frame layout in a few lines (sync, length, checksum, typed fields with scale, unit, enums, bits); the consoles (right-click > Decode protocols), the Line Monitor and the plotter then show every field.
- **Tools > Lua Scripts...**: tests and automation in Lua 5.4 (spm.send, spm.expect, spm.check... with PASS / FAIL, also `--lua` without windows); a script as the hook of a tab (console right-click > Lua hook); Lua dissectors and plotters in library.lua.
- **Tools > Serial Plotter...** (Ctrl-L): draws the numbers a device prints, one line per sample (`21.5,40` or `temp:21.5 hum:40`), one colour per value; **Pause**, scale, **Export CSV...**.
- **Tools > Test Sequences...** (Ctrl-T): a script - `send "AT\r\n"`, `expect "OK" within 500`, `delay 100`, `repeat 3` ... `end` - run against the device; every expectation PASS / FAIL, the run **PASSED** or **FAILED**. Also `set` / `add` variables, `crc`, `expect regex`, `expect none`, `goto`; and without windows: `SerialPM.exe --run test.spmseq --port COM3 --junit report.xml` (exit code 0 = passed).
- **File > Export Profile...** / **Import Profile...**: the whole setup of the ports (settings, automation, receive options, commands, highlights) in a file.

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10. License

- The 14-day trial includes every feature. Afterwards, without a license, the program still opens ports and shows and logs everything received (and the Byte Analyzer and Protocols window keep working); sending (also commands, replay, the Modbus master and test sequences), auto replies, periodic messages, file transfer and forwarding need a license.
- Licenses are per PC. Open **Help > License...** (Ctrl-K): it shows the state, the features and **This PC's unique ID (UID):** with **Copy**.
- To get a license, send the UID - or the request file from **Save Request...** (also in the data folder as license-request.txt) - to the license issuer.
- Install the license you receive with **Load License...**, **Paste License**, or by dropping the file on the License window. **Remove License** keeps a copy as license.lic.bak.

11. Help and support

- **Help > User Manual** (F1): the complete manual, in the language in use. **Help > Quick Guide**: this guide. Both are PDF files in the **manuals** folder next to the exe.
- **Help > Version Info...**: version, build, data folder and version history. **Help > About...**: tabs with the program, the developer (report a problem by e-mail), the libraries, the ports of this PC, the license, the changes and the system details.
- For support, set **Settings > Log Level...** to Debug, repeat the problem and send log.txt from the data folder (**Settings > Open Data Folder**) to the licensor, Fation Coga, together with the version shown in **Version Info...**.

12. Legal notices

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